



IIT KHARAGPUR
SCSQR
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RELIABILITY DEMONSTRATION OF PUSH BUTTONS

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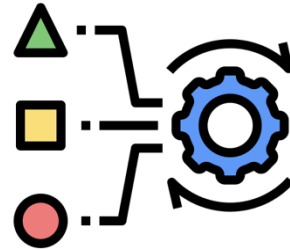
SCOPE



Objective



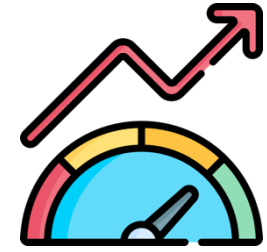
Introduction



Test operating
conditions



Test Setup



Accelerated life testing



Test procedure
Binomial sequential test plan
Test observations
Summary of observations

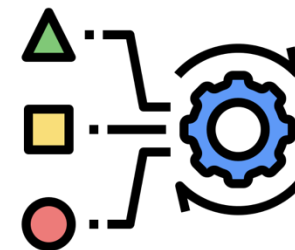
PART #1



Objective



Introduction



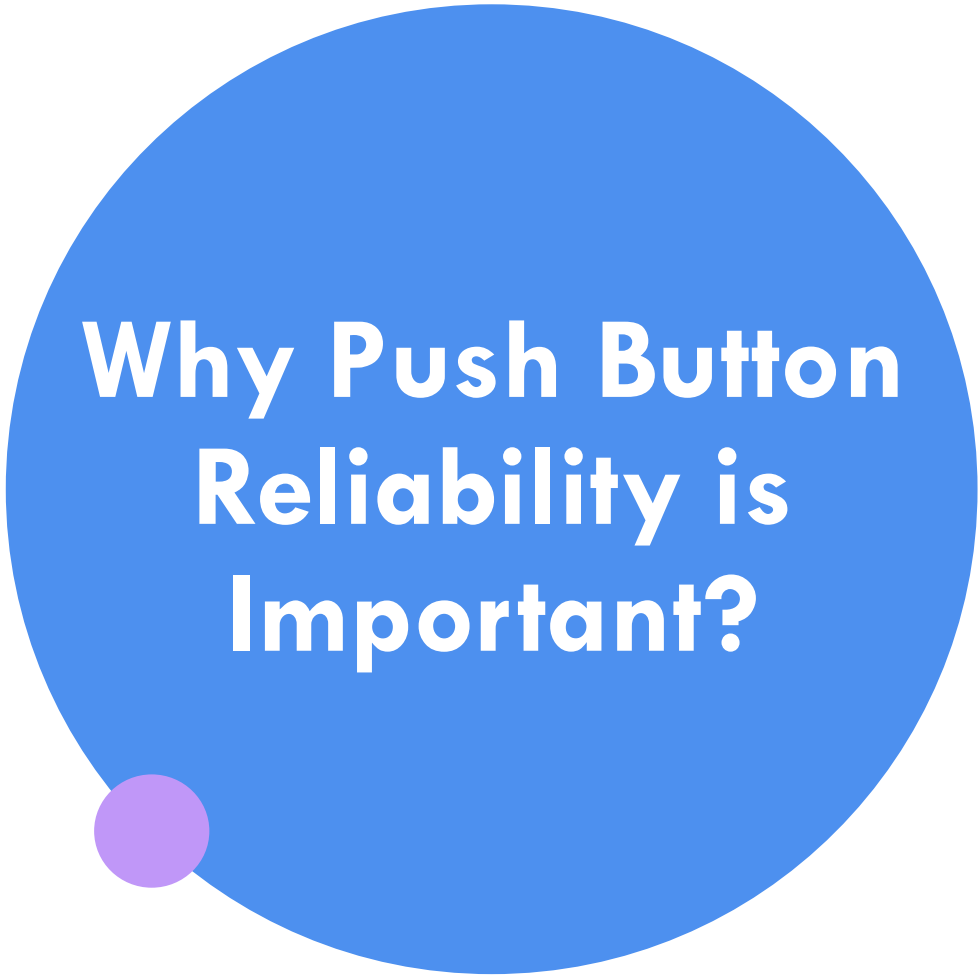
Test operating conditions



Test Setup



Accelerated life testing



Why Push Button Reliability is Important?

Push buttons are widely used in safety-critical and industrial applications:

- Industrial control panels
- Railway signalling systems
- Medical devices
- Automotive dashboards Emergency shutdown systems

Failure of push buttons can lead to:

- System malfunction
- Safety hazards
- Production downtime
- Increased maintenance cost



OBJECTIVE



To demonstrate the minimum acceptable reliability of push button switches for a specified operating time with given Producer's and Consumer's Risk

INTRODUCTION

KEY FACTORS IN RELIABILITY DEMONSTRATION

Define

- Define R_u
- Define R_L

Risk

- Set Producer's risk (α)
- Set Consumer's risk (β)

Scale

- Sample size
- Total test time

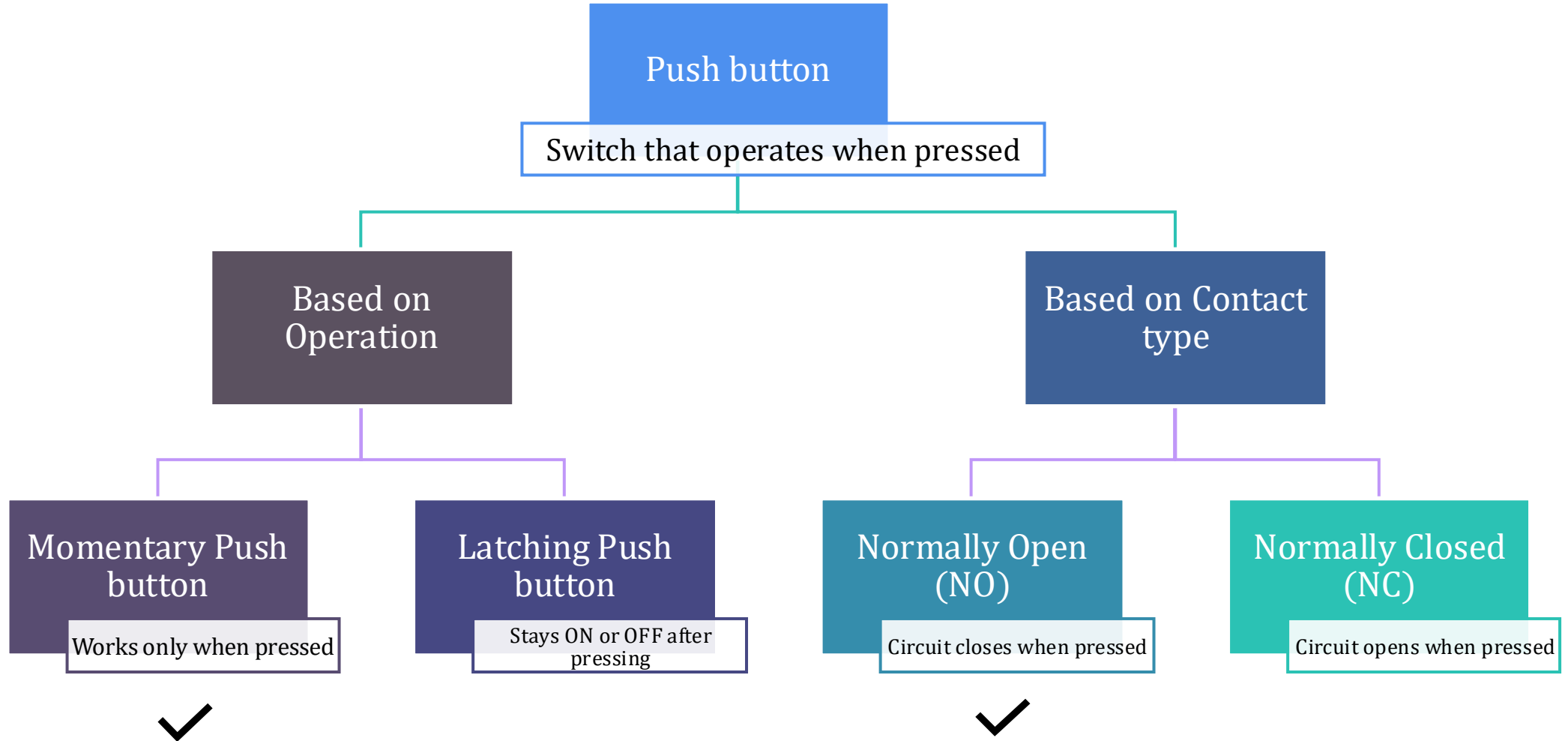
Environment

- Choose normal operating conditions or
- Accelerated testing

Method

- Pick test design

TYPES OF PUSH BUTTONS



PUSH BUTTON SPECIFICATIONS AND OPERATING CONDITIONS

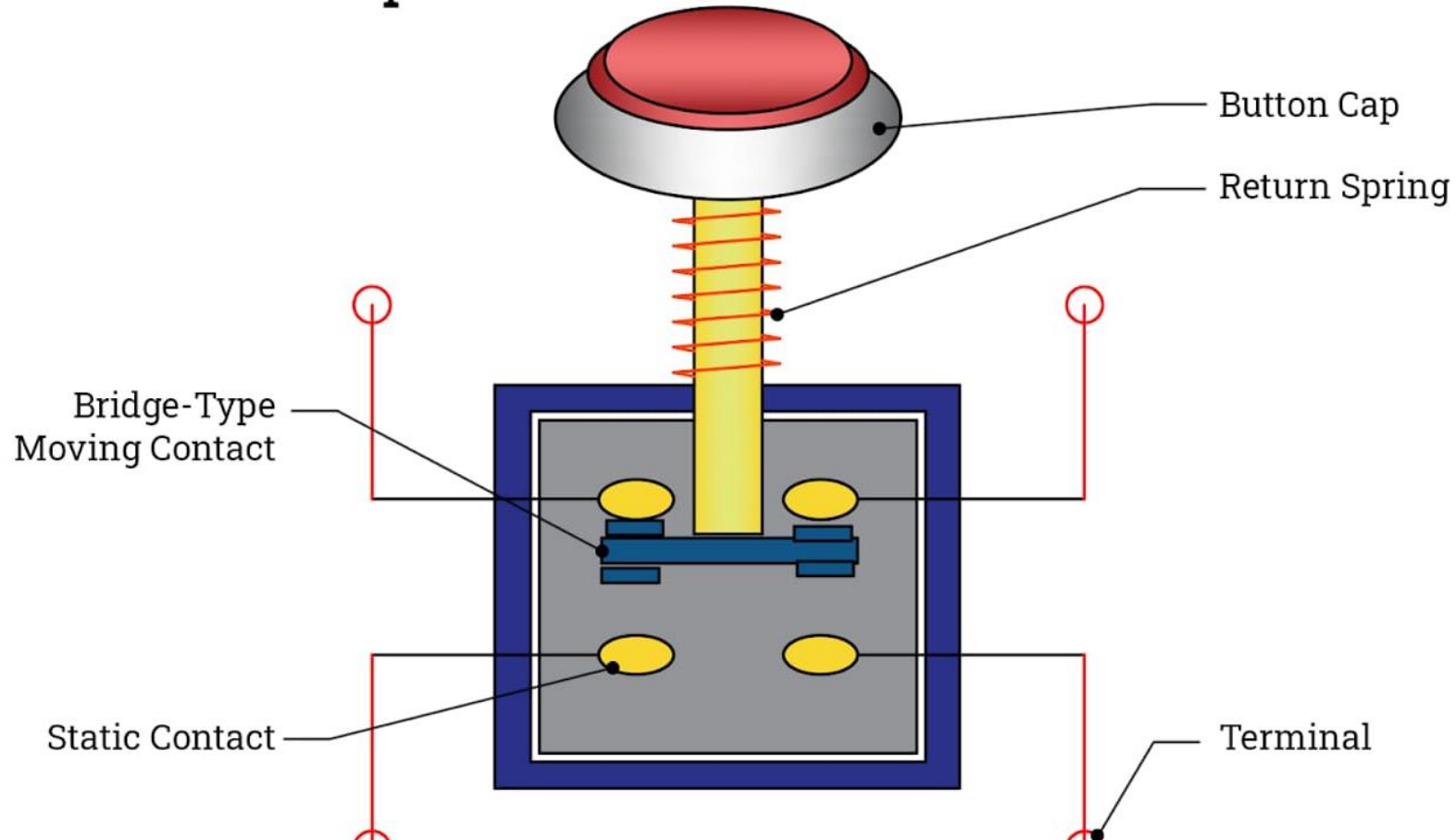
Parameters / specifications	Description
Switch Type	Push Button Momentary, Roundhead
Terminal Style	Screw type
Operating Temp.	10°C to 80°C
Mounting	Panel, screw type
Current & Voltage Rating	5A, 220 V AC



**SPST PUSH BUTTON -
SE952 PUSH TO ON
MEDIUM**

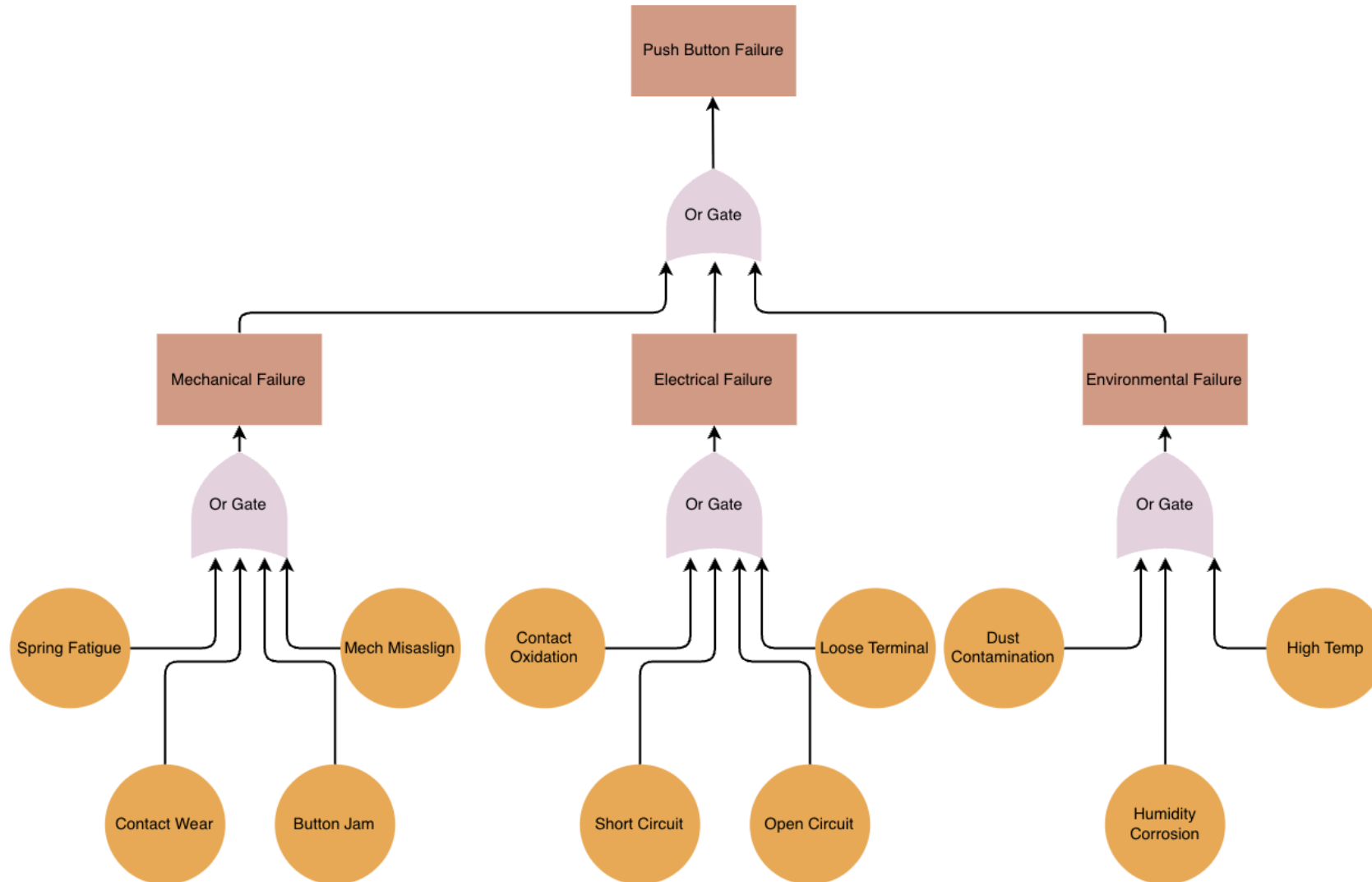
TEST OPERATING CONDITIONS	
Electrical Supply	220 V AC supply
Temperature	25 C
Cycles per minute	30
Relative Humidity	70 – 80 % (approx.)

The Operation of a Pushbutton Switch

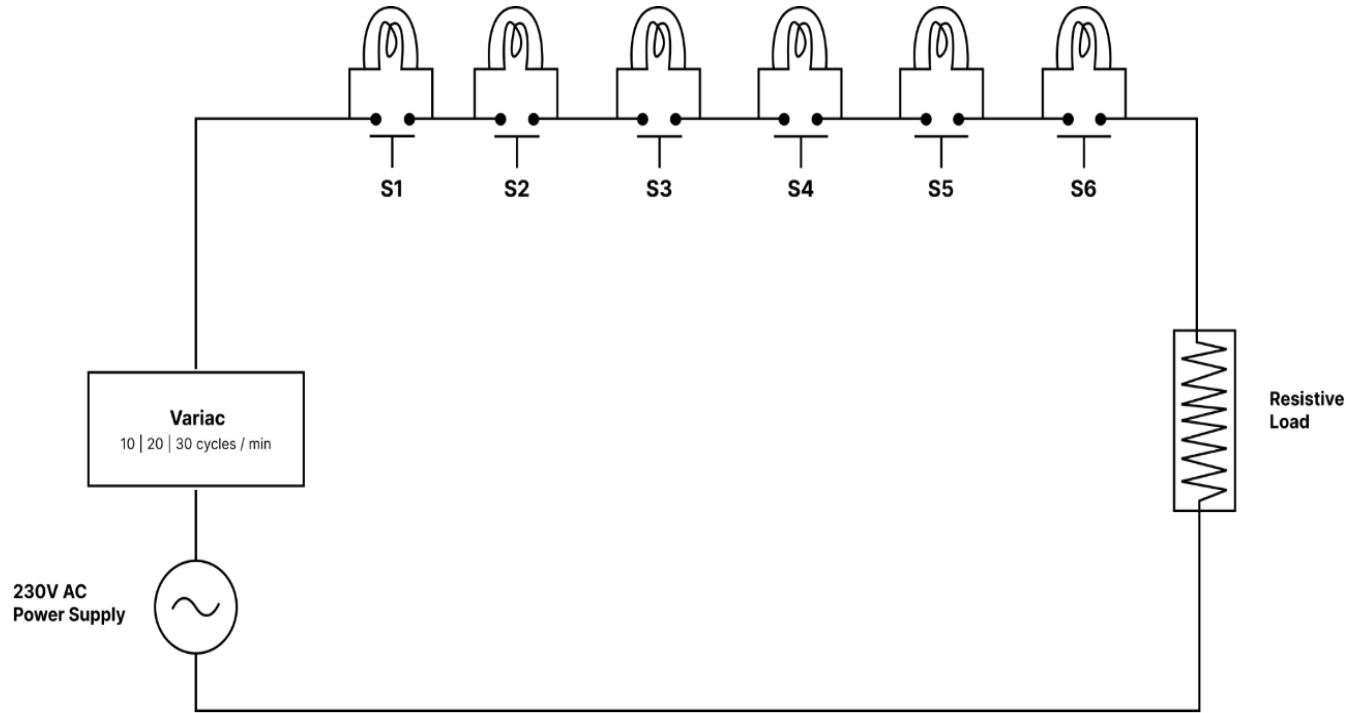


**A QUICK RECAP TO
RELIABILITY
ENGINEERING LAB
OUTPUTS**

FAULT TREE DIAGRAM FOR A PUSH BUTTON FAILURE



TEST SETUP



Circuit Diagram of the test setup

- The circuit was connected as shown in the circuit diagram above. The circuit is designed in such a way that the failure of the switches can be accelerated by either increasing the rate of switching speed (on - off cycle) in rated load conditions or by increasing the load.



Actual Test Setup

RESULTS FROM RT LAB - ALT



Accelerated stress (A)	Shape parameter (β)	Scale parameter (θ)
0.8	2.6063	16974.67
1	2.6017	8409.926
1.5	3.9023	5621.732

$$\begin{aligned}\beta_{\text{avg}} &= \frac{\beta_{0.8A} + \beta_{1A} + \beta_{1.5A}}{3} \\ &= \frac{2.6063 + 2.6017 + 3.9023}{3} \\ &= 3.036767\end{aligned}$$

Using Inverse Power Law:

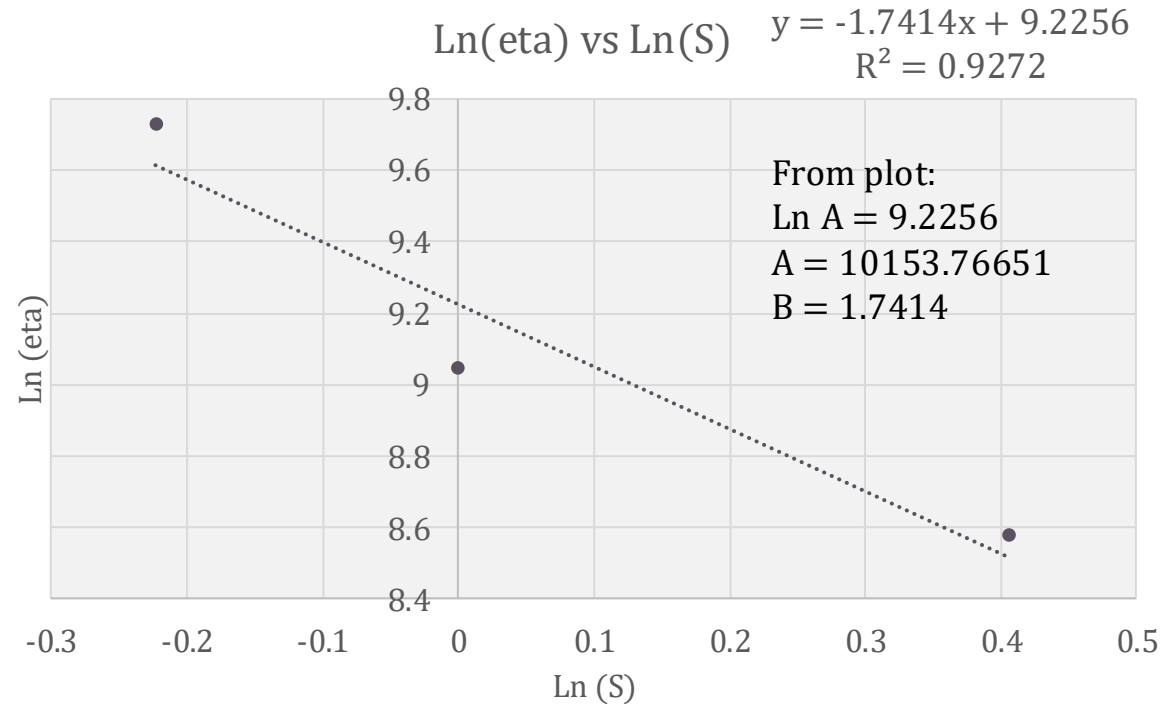
$$L = \frac{A}{S^B}$$

$$\ln L = \ln A - B \ln S$$

$$\theta = \left[\frac{1}{N} \sum_i^N t_i^{\beta} \right]^{\frac{1}{\beta}}$$

Stress (S)	Beta avg (β_{avg})	Theta (θ)	Ln eta ($\ln \theta$)	Ln S
0.8	3.036767	16787.26	9.728376	-0.22314
1	3.036767	8505.405	9.048457	0
1.5	3.036767	5336.599	8.582344	0.405465

RESULTS FROM RT LAB - ALT (Contd.)



- Electrical life = 30000 cycles
- Life (L) = $\theta = \frac{A}{S^B}$
- Acceleration Factor (AF) = $\frac{L_{\text{nominal}}}{L}$
- MTTF = $\theta * \Gamma(1 + \frac{1}{\beta})$
- $t(\text{test}) = \text{required testing time or cycles}$

$$= \frac{\text{Electrical life}}{\text{AF}}$$
- $R_U = e^{-\left(\frac{t}{\theta}\right)^{\beta_{avg}}}$

Conditions	Stress (S)	Life (theta)	AF	MTTF	t (test)
Nominal	0.2	167422.39	1	149585.8	30000
case 1	0.8	14975.67	11.17962584	13380.21	2683.452955
case 2	1	10153.767	16.48869807	9072.019	1819.428064
case 3	1.5	5011.6629	33.40655439	4477.738	898.0273646

Parameters	Value
Electrical life (cycles)	30000
Nominal stress (A)	0.2
theta	167422.3902
Reliability (Ru)	0.994613614



RELIABILITY DEMONSTRATION TESTS

RELIABILITY DEMONSTRATION TESTS

Binomial Test

- Sample size fixed before hand
- Allowed no of failures decided before hand

Example:

Test 50 push buttons

Accept product if failures ≤ 2

- Decision is made only **after all tests are completed**
- For high reliability components like push buttons, failures are rare, so testing many samples becomes inefficient

Sequential Test

- Testing is done step by step
- After each trial, a decision to accept/reject/ continue taken

Advantages

- Requires few samples
- Reduces cost
- Efficient Method

BINOMIAL DEMONSTRATION PLAN (COMPARISON)

Good product should be accepted with probability $\geq 1 - \alpha$

$$\sum_{X=0}^C {}^nC_x (1-R_U)^x R_U^{n-x} \geq 1 - \alpha$$

Bad product should be accepted with probability $\leq \beta$

$$\sum_{X=0}^C {}^nC_x (1-R_L)^x R_L^{n-x} \leq \beta$$

Parameters	Values	No. of Failures (x)	Total Sample Size (n)
Desired Reliability, R_U	0.99	0	$n \leq 22$ $n \geq 31$
Min Reliability, R_L	0.95		
Producer's Risk, α	0.2	1	$n \sim 59$
Consumer's Risk, β	0.2		

ASSUMPTIONS

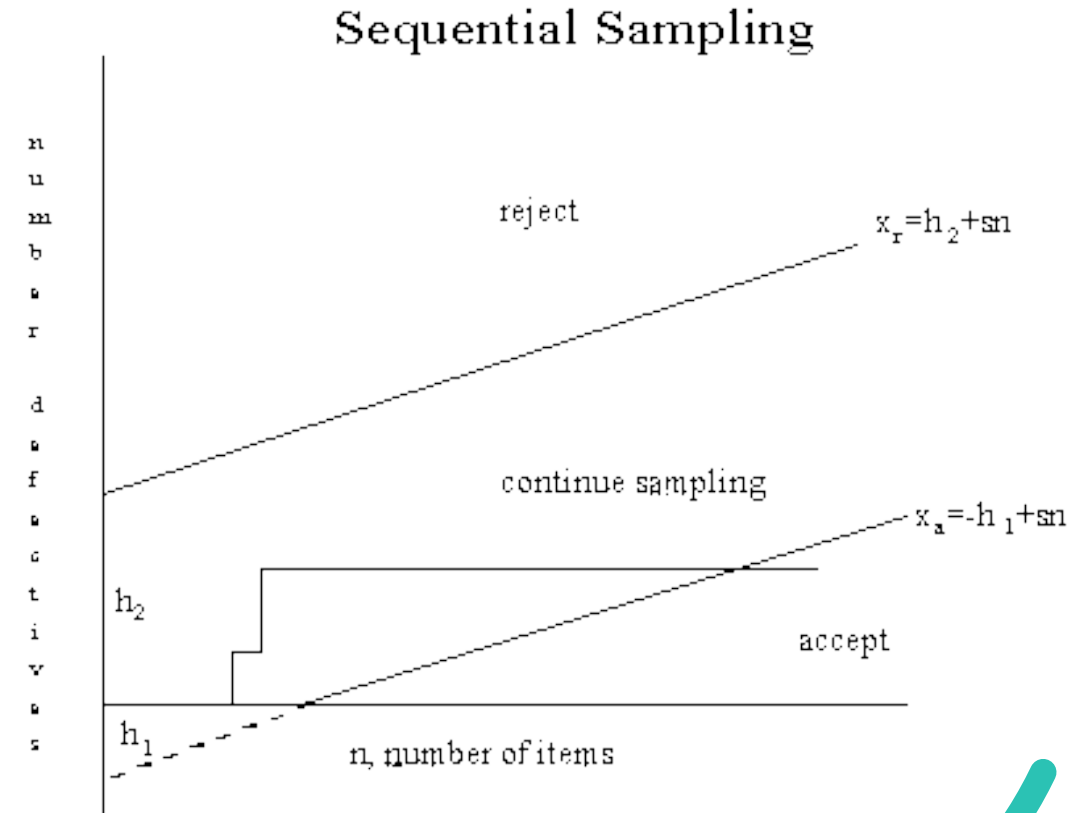
- All switches are from same manufacturer/lot/batch
- Test Equipments are serviceable
- Failure mechanism same at normal and accelerated stress
- Samples are identical and independent
- Operating conditions remain constant
- Pass/fail criteria clearly defined
- No external disturbances (vibrations)



- Test procedure
- Sequential test plan
- Test observations
- Summary of observations

TEST PROCEDURE

- A Sequential Test Plan is selected for this analysis due to its efficiency in evaluating the system's reliability by minimizing the average sample size while maintaining statistical rigor.
- The term sequential indicates that this type of test is accomplished by conducting a succession, or a sequence of runs on one or more devices.
- One of the following three decisions is made after each test run,
 - **Accept:** The device has demonstrated the reliability requirement.
 - **Continue Testing:** The status of the device's reliability is undetermined.
 - **Reject:** The device has failed to demonstrate the reliability requirement



SEQUENTIAL TEST PLAN

- To use the sequential testing analysis technique, four quantities need to be specified R_U , R_L , α and β .
- Two additional terms used here are: $Q_L = 1 - R_U$ and $Q_U = 1 - R_L$

Parameters required	Description
R_U	• Design objective for the reliability level • Reliability obtained from ALT
R_L	• Minimum acceptable reliability level to the customer. • Reliability we want to demonstrate
α	• Producer's risk • Risk of having equipment that has a true reliability equal to or greater than the contract's design objective rejected
β	• Consumer's risk • Risk of having equipment that has a true reliability less than the minimum required gets accepted

- The consumer's risk, β , is associated with the minimum acceptable reliability, R_L
- The producer's risk α , is associated with the design objective reliability, R_U .

SEQUENTIAL TEST PLAN (Contd.)

The following step-by-step method may be used to construct a sequential testing plan to demonstrate that a device meets or fails to meet a specified, acceptable minimum reliability, R_L .

- 1) Define R_U and R_L
- 2) Define α and β
- 3) Determine the value of the accept intercept.

$$h_0 = \frac{\ln\left(\frac{\beta}{1-\alpha}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)}$$

- 4) Determine the value of the reject intercept.

$$h_1 = \frac{\ln\left(\frac{1-\beta}{\alpha}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)}$$

- 5) Determine the value of the slope of the test plan lines.

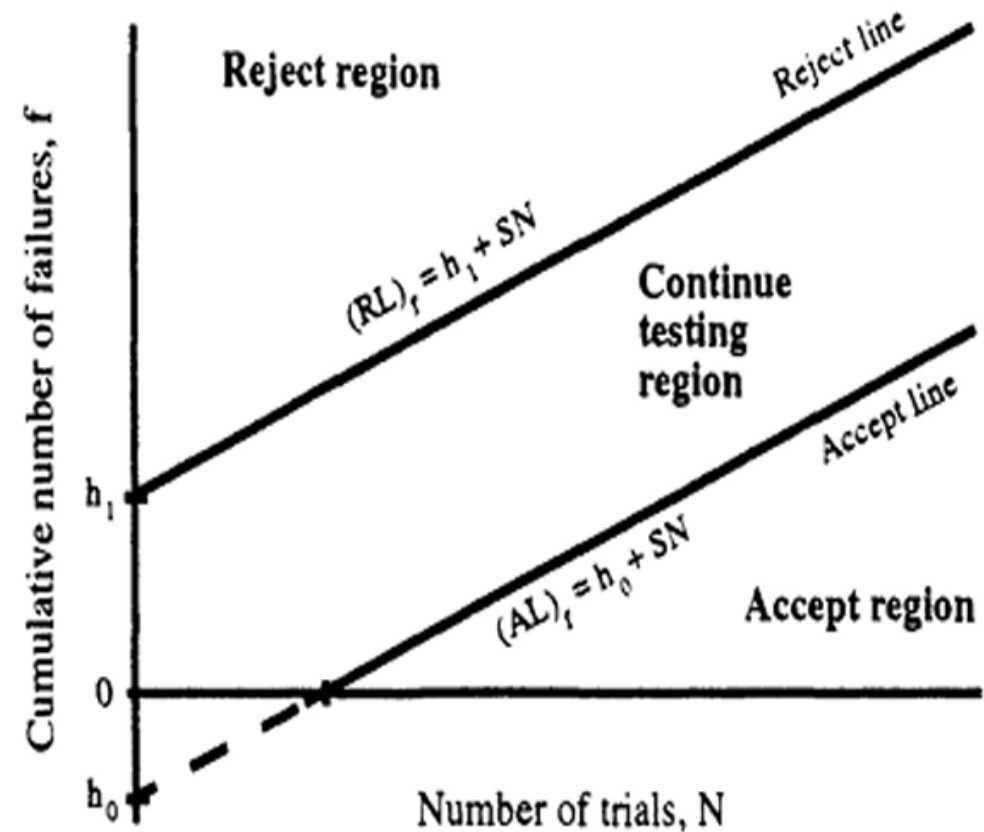
$$S = \frac{\ln\left(\frac{R_U}{R_L}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)}$$

- 6) Determine the equations for the accept line,

$$AL_f = h_0 + SN$$

- 7) Determine the equations for the reject line,

$$RL_f = h_1 + SN$$





TEST OBSERVATIONS

- $R_U = 0.99$ and $R_L = 0.95$ (consumer grade & industrial grade)
- $Q_L = 1 - 0.99 = 0.01$ and $Q_U = 1 - R_L = 1 - 0.95 = 0.05$
- $\alpha = 0.2$ and $\beta = 0.2$

$$h_0 = \frac{\ln\left(\frac{\beta}{1-\alpha}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)} = \frac{\ln\left(\frac{0.2}{1-0.2}\right)}{\ln\left(\frac{0.05}{0.01}\right) - \ln\left(\frac{0.95}{0.99}\right)} = -0.83983$$

$$h_1 = \frac{\ln\left(\frac{1-\beta}{\alpha}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)} = \frac{\ln\left(\frac{1-0.2}{0.2}\right)}{\ln\left(\frac{0.05}{0.01}\right) - \ln\left(\frac{0.95}{0.99}\right)} = 0.83983$$

$$S = \frac{\ln\left(\frac{R_U}{R_L}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)} = \frac{\ln\left(\frac{0.99}{0.95}\right)}{\ln\left(\frac{0.05}{0.01}\right) - \ln\left(\frac{0.95}{0.99}\right)} = 0.025$$

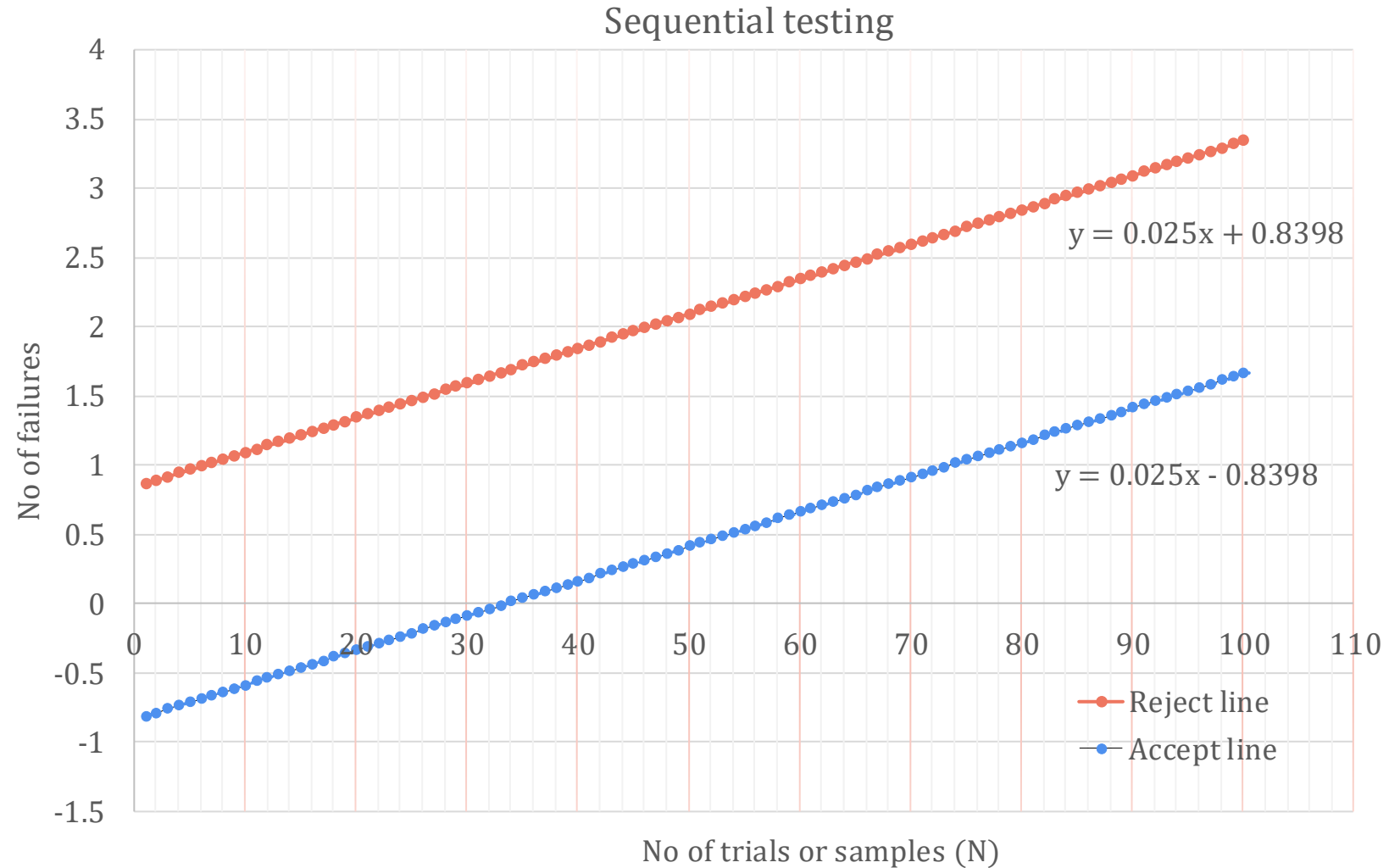
- Accept line : $AL_f = h_0 + SN = -0.83983 + 0.025 N$
- Reject line : $RL_f = h_1 + SN = 0.83983 + 0.025 N$

N	reject line	accept line
1	0.864817	-0.814846
2	0.889803	-0.789861
3	0.914788	-0.764876
4	0.939774	-0.73989
5	0.964759	-0.714905
6	0.989744	-0.689919
7	1.01473	-0.664934
8	1.039715	-0.639948
9	1.064701	-0.614963
10	1.089686	-0.589978
.	.	.
.	.	.
.	.	.



TEST OBSERVATIONS (Contd.)

- Minimum number of trials for an accept decision ($AL_f = 0$)
 $AL_f = h_0 + SN$
 $0 = -0.83983 + 0.025 N$
 $N = 34 \text{ samples}$
 $= (6, 6, 6, 6, 6, 4)$
- Testing cycles for reliability demonstration (T)
 $= \text{Electrical life} / \text{AF}$
 $= 30,000 / 11.18$
 $= 2683 \text{ cycles}$
 $\approx 2700 \text{ cycles}$



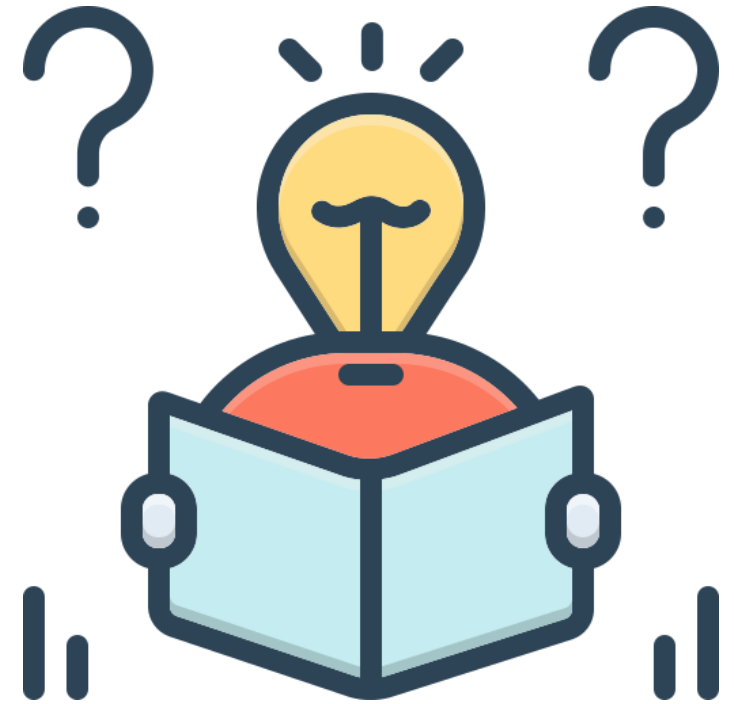


TEST OBSERVATIONS (Contd.)

Sample No.	TTF (Cycles)	Sample No.	TTF (Cycles)	Sample No.	TTF (Cycles)
1	All push buttons operated satisfactorily upto 2700 cycles	13	All push buttons operated satisfactorily upto 2700 cycles	25	All push buttons operated satisfactorily upto 2700 cycles
2		14		26	
3		15		27	
4		16		28	
5		17		29	
6		18		30	
7		19		31	
8		20		32	
9		21		33	
10		22		34	
11		23		X	
12		24		X	

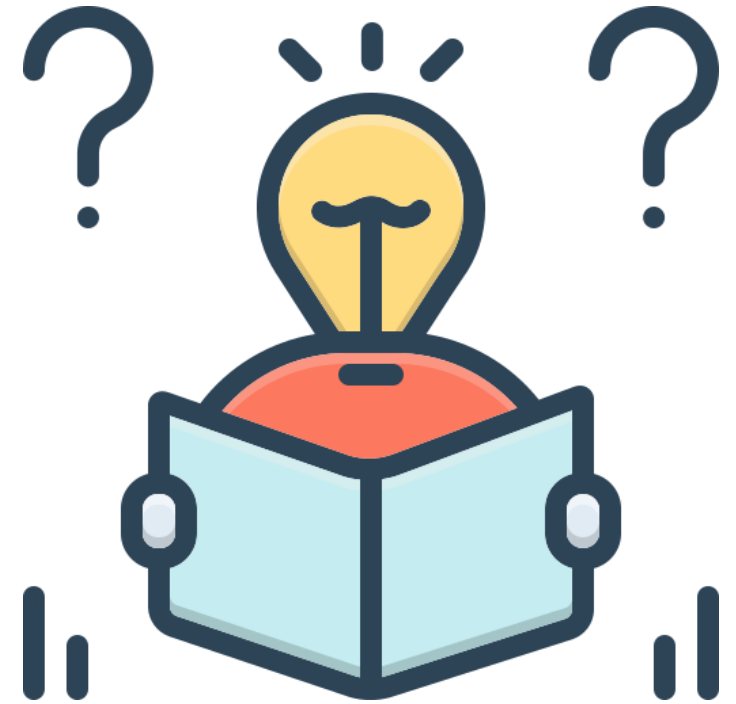
CONCLUSION

- From the sequential testing plot, it is observed that the acceptance line intersects the trial axis at 34, **indicating that acceptance is achieved with 0 failures up to 34 samples.**
- The absence of failures during the test till 2700 cycles indicates that the push buttons exhibit high reliability under the given operating conditions.
- The product satisfies the required reliability limits ($RL = 0.95$) within the specified producer's and consumer's risk levels.
- The sequential testing method efficiently validates reliability using a minimum number of samples, thereby reducing testing time.
- Hence, the push buttons are considered reliable and suitable for practical applications within the specified limits.



FINAL RELIABILITY STATEMENT

The push button switches exhibited a min Reliability of 0.95 for 2700 cycles with Producer's and Consumer's Risk at 20% under stress current of 0.8A





THANK YOU